

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.599934218386754, median 0.4310296636116538, std: 0.5396445322
Reprojection error (cam1): mean 0.6112779388443921, median 0.47977893472757493, std: 0.48980230
Gyroscope error (imu0): mean 2.369234829013399, median 1.7365737779162447, std: 1.88661311106
Accelerometer error (imu0): mean 0.13574128663333646, median 0.11150396405522638, std: 0.0974199

Residuals

Reprojection error (cam0) [px]: mean 0.599934218386754, median 0.4310296636116538, std: 0.5396445
Reprojection error (cam1) [px]: mean 0.6112779388443921, median 0.47977893472757493, std: 0.48980
Gyroscope error (imu0) [rad/s]: mean 0.33506040276374494, median 0.24558861887906364, std: 0.2668
Accelerometer error (imu0) [m/s^2]: mean 0.19196716853083817, median 0.15769041822526325, std: 0.1

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[ 0.99987957 -0.0106826 -0.0112573  0.01832679]
 [ 0.01057217  0.99989585 -0.00982423  0.00864658]
 [ 0.01136107  0.00970404  0.99988837  0.08075767]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam0 to imu0):

```
[[ 0.99987957  0.01057217  0.01136107 -0.01933349]
 [-0.0106826  0.99989585  0.00970404 -0.00923358]
 [-0.0112573 -0.00982423  0.99988837 -0.0804574 ]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.008901881574933197

Transformation (cam1):

T_ci: (imu0 to cam1):

[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):

[[0.99979642 0.01000066 0.01752434 0.07598161]
[-0.01017159 0.9999013 0.00969211 -0.01011996]
[-0.01742568 -0.00986839 0.99979946 -0.08206679]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)

-0.00882664858323035

Baselines:

Baseline (cam0 to cam1):

[[0.99998084 0.00057068 -0.00616368 -0.09533276]
[-0.000571 0.99999984 -0.00005059 -0.0000828]
[0.00616365 0.00005411 0.999981 -0.00005267]
[0. 0. 0. 1.]]

baseline norm: 0.09533281124934556 [m]

Gravity vector in target coords: [m/s^2]

[-0.18903579 -9.55838677 -2.18401711]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [430.7661746442717, 430.850377143008]

Principal point: [424.8191678739607, 239.73084864213763]

Distortion model: radtan

Distortion coefficients: [-0.004611570913312722, -0.0009360260907051189, -0.0009333871766183999, -

Type: aprilgrid

Tags:

Spacing 0.0075 [m]

cam1

Camera model: pinhole

Focal length: [430.58624390582474, 431.0449770712561]

Principal point: [427.61126948415335, 239.86900817185756]

Distortion model: radtan

Distortion coefficients: [-0.004174585300012547, -0.00013609420224485136, -0.0007918238646686342,

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.025 [m]

Spacing 0.0075 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 200.0

Accelerometer:

Noise density: 0.1

Noise density (discrete): 1.4142135623730951

Random walk: 4e-05

Gyroscope:

Noise density: 0.01

Noise density (discrete): 0.1414213562373095

Random walk: 2e-06

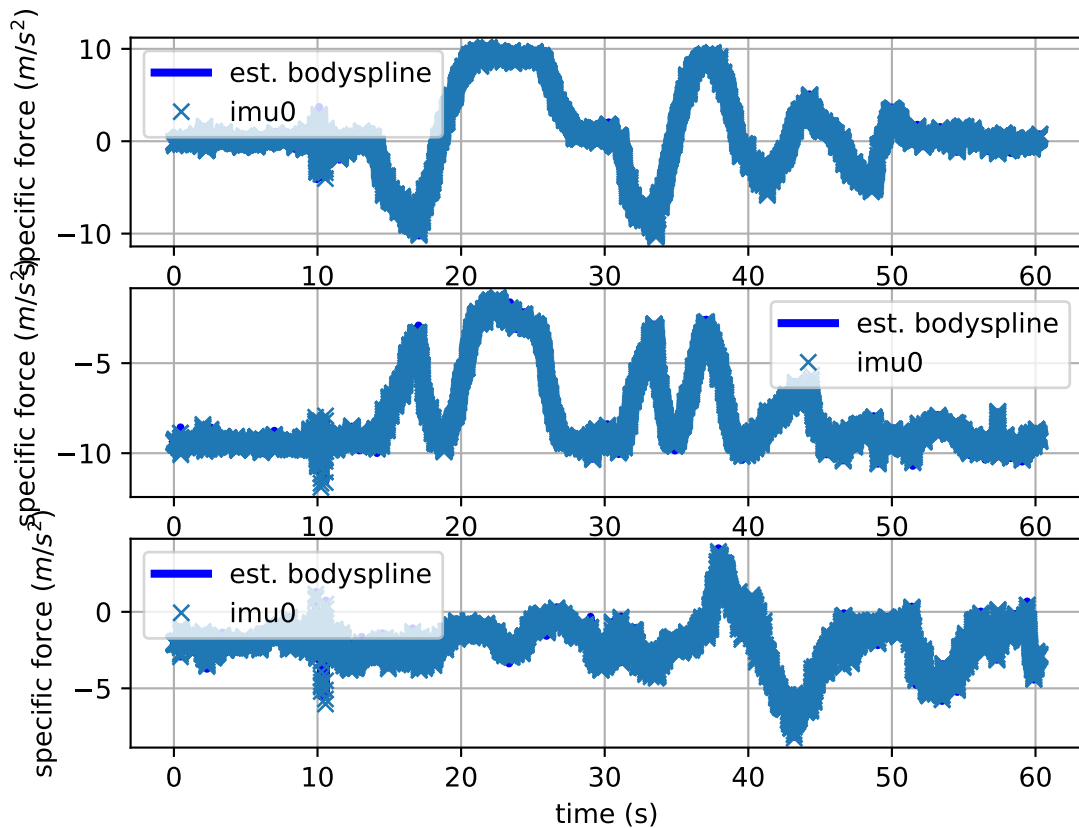
T_i_b

[[1. 0. 0. 0.]

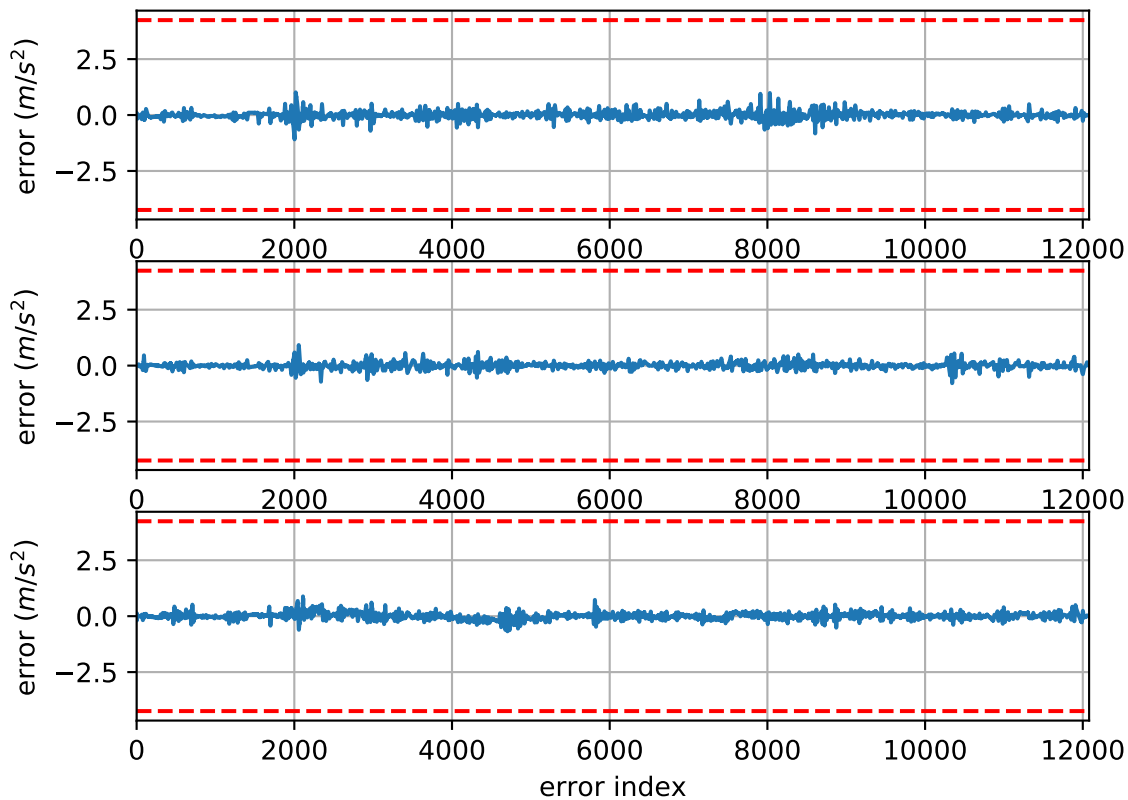
[0. 1. 0. 0.]

[0. 0. 1. 0.]

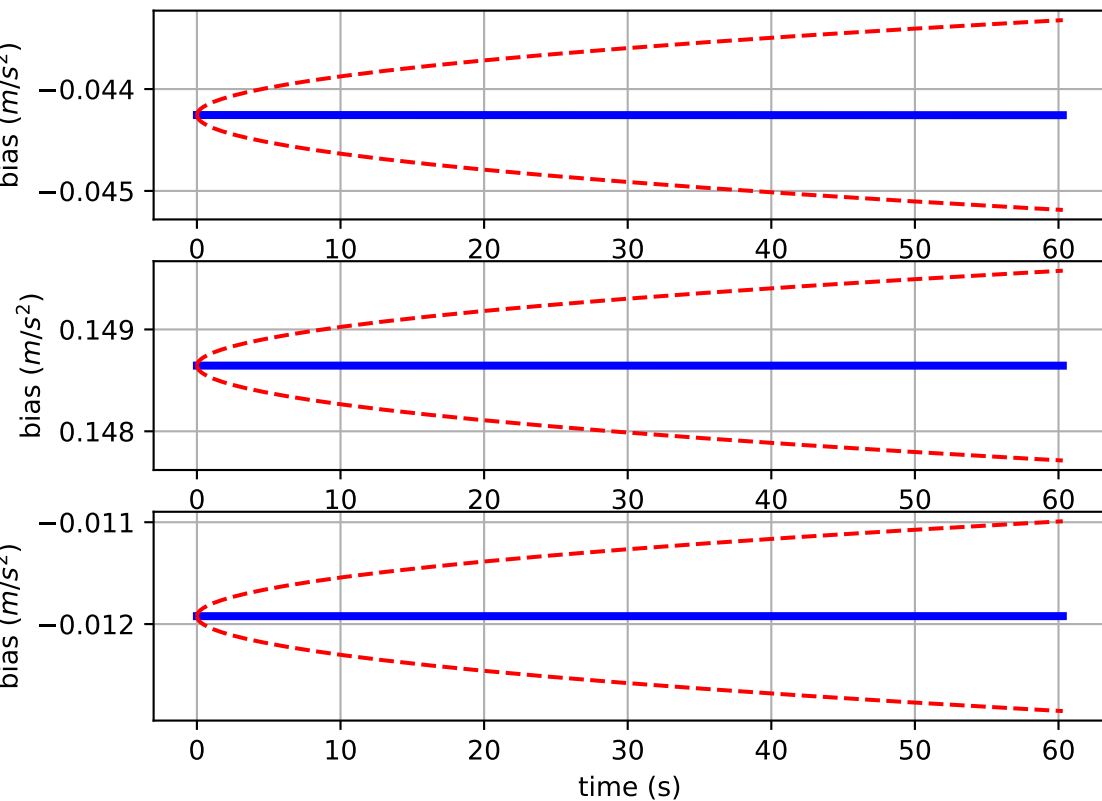
Comparison of predicted and measured specific force (imu0 frame)



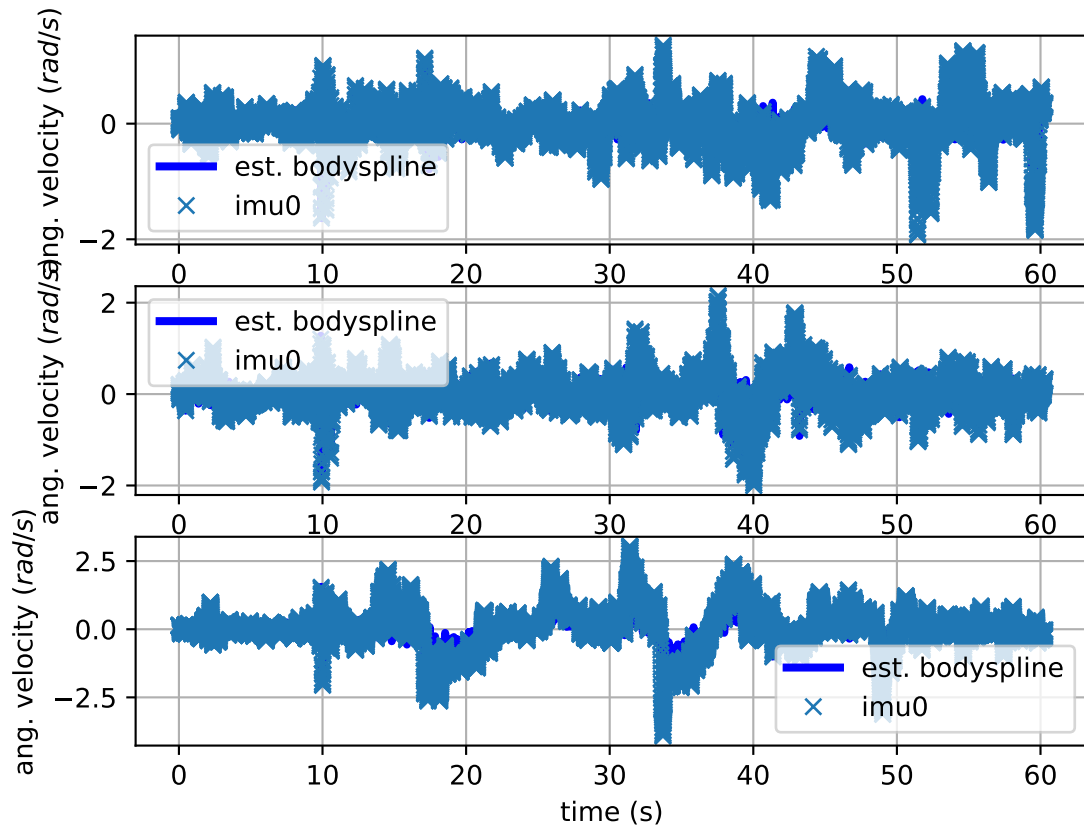
imu0: acceleration error



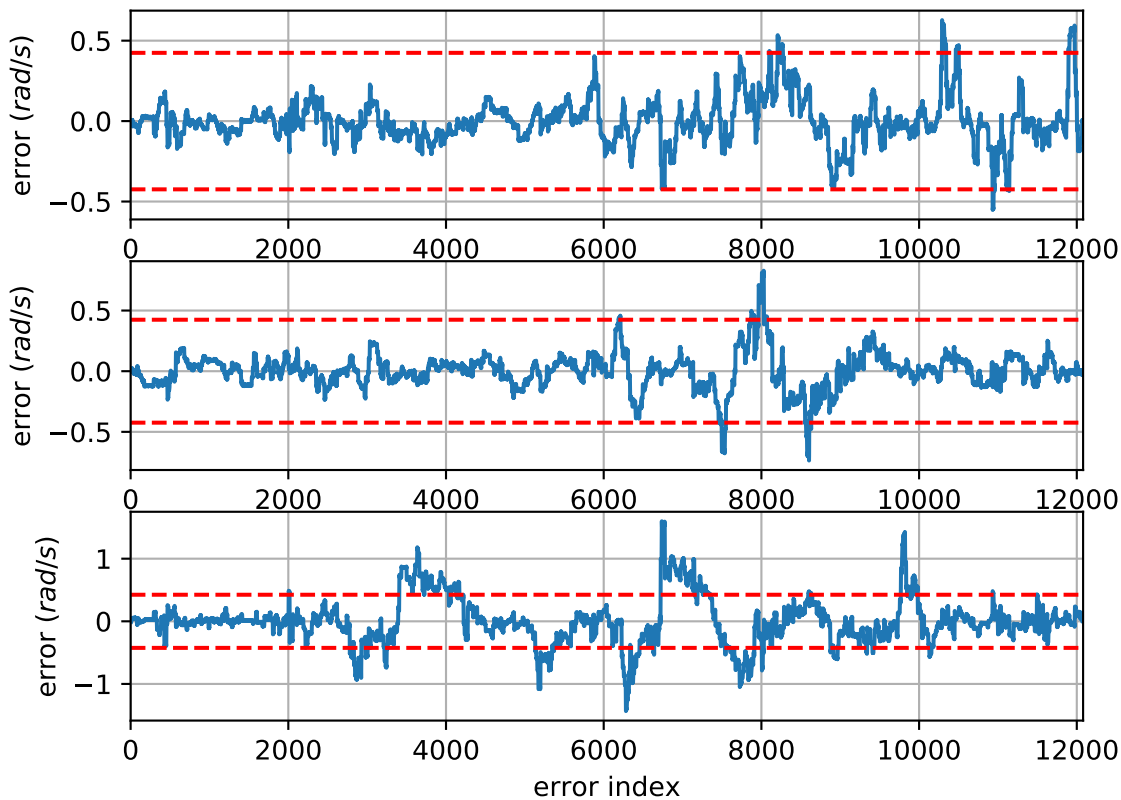
imu0: estimated accelerometer bias (imu frame)



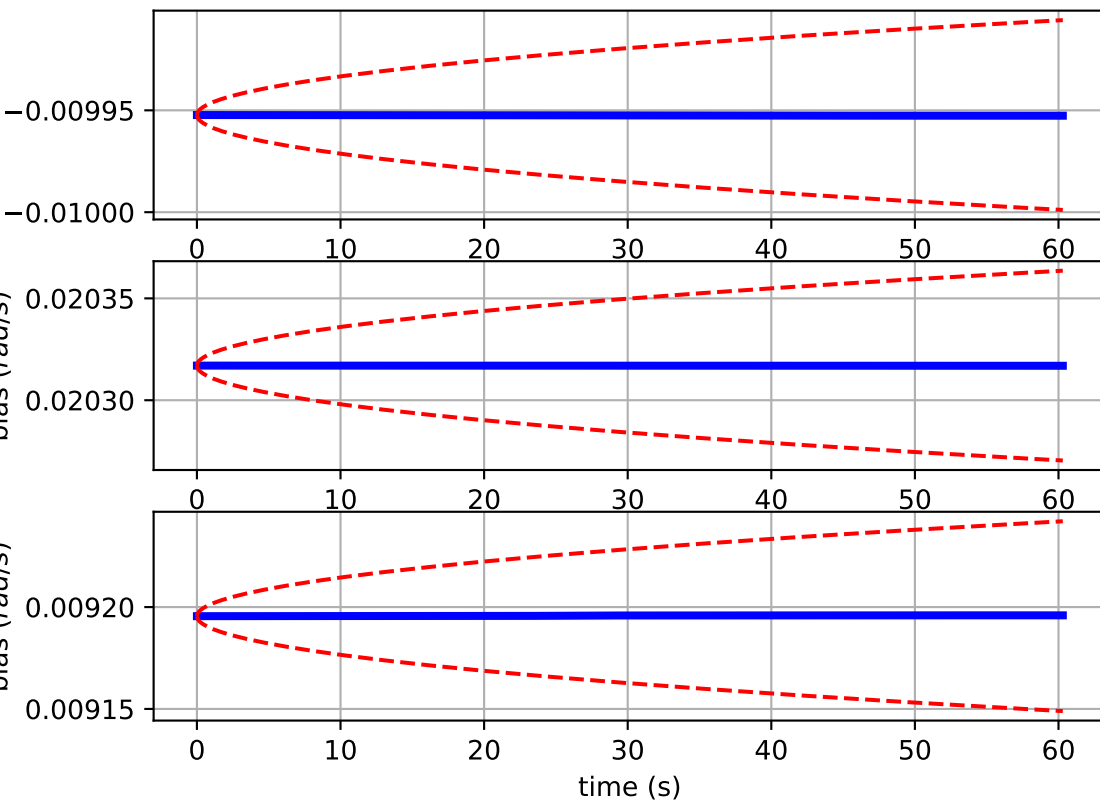
Comparison of predicted and measured angular velocities (body frame)



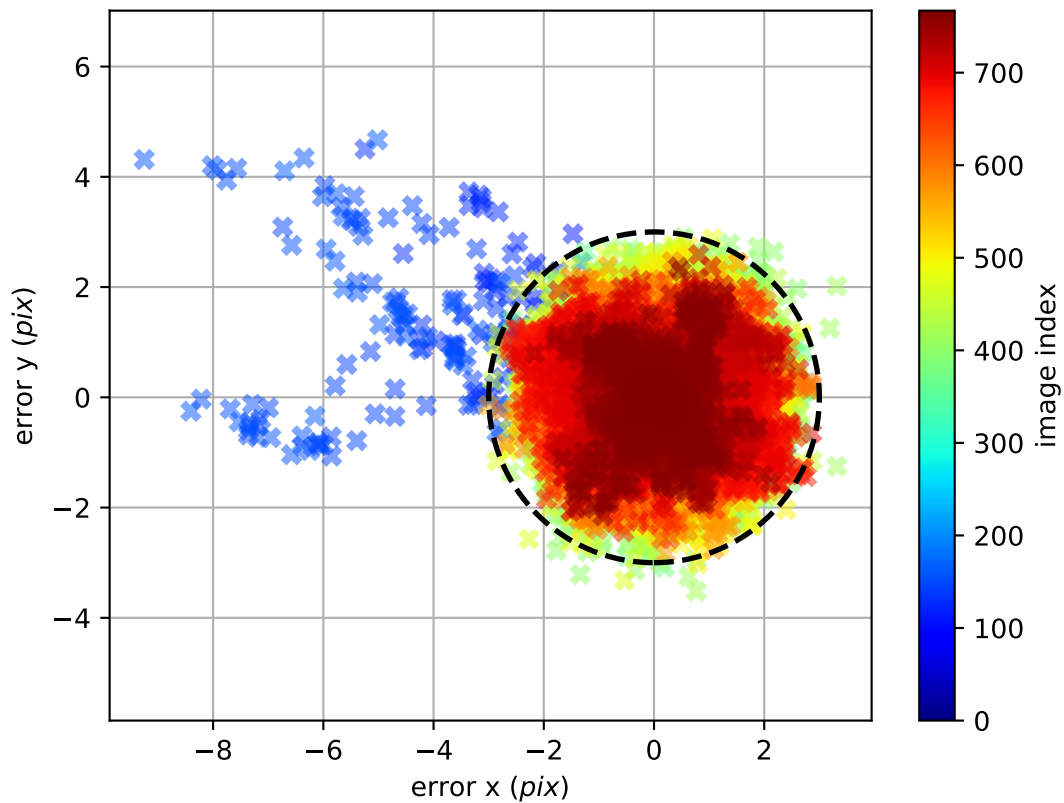
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

